

Light -activation sources

Light activation sources for composite resin filling materials are:

1. Ultraviolet light source:
2. Visible light sources:
 - A. Quartz-Tungsten-halogen (QTH) light source:
 - B. Argon laser lights
 - C. Short-Arc Xenon Sources (Plasma-Arc Curing lights (PAC))
 - D. Blue light- emitting diode curing units (LED) s.

Light curing techniques for composite resin

1. Continuous curing: a light of constant intensity is applied to the composite resin for a specific period of time.
2. Ramp Cure: light is initially applied at low intensity and gradually increased over time to high intensity levels.
3. Pulse Delay Cure.
4. Soft-cure technique.

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Polymerization:

Resin composite restoratives solidify by means of the chemical process termed polymerization. The polymerization of the resin matrix produces a gelation in which the restorative material is transformed from a viscous-elastic into a rigid-plastic phase. During the early stages of polymerization, monomers are mainly converted into polymeric chains.

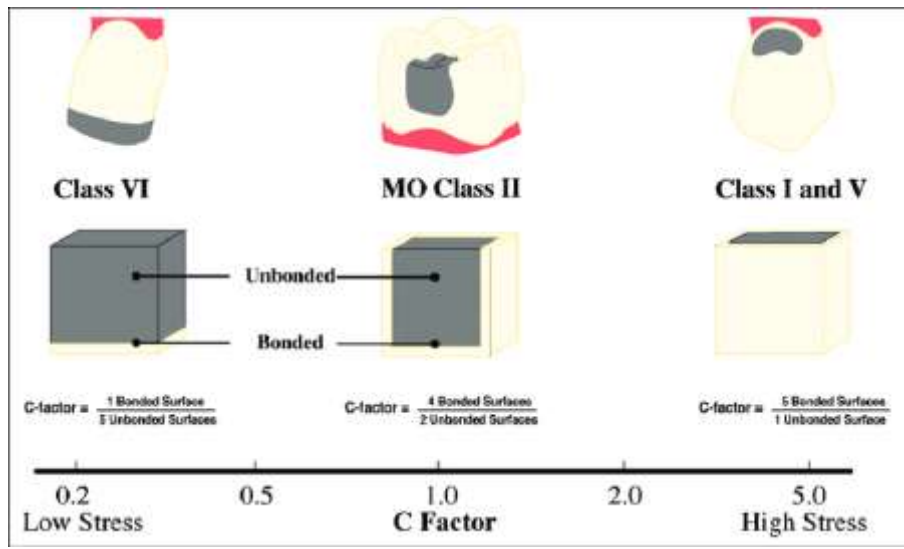
After a certain degree of conversion has been attained, the predominant reaction is the cross-linking of the polymeric chains, resulting in a strong polymeric network.

Factors affecting polymerization shrinkage stress:

1-Factors related to the cavity design:

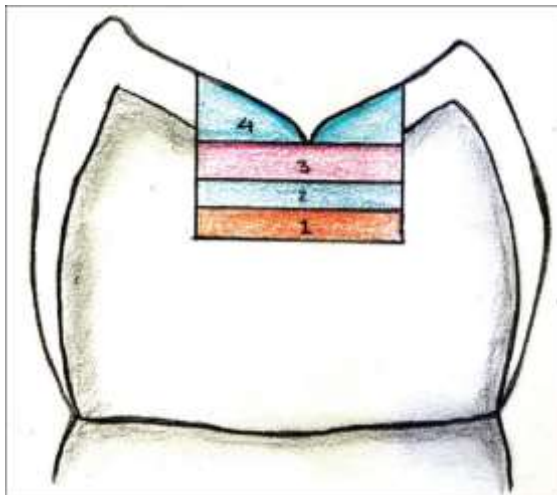
Stress developed during curing can be minimized by consideration of the ratio between the bonded and unbonded surface area (c- factor). When this ratio increases, as in Class I and Class V situations, increase the shrinkage stress loading on the tooth- resin interface leading to de bonding.

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2-Factors related to the placement technique:

The second factor that might reduce polymerization shrinkage is to insert resin composites in increments to reduce the volume of the resin that is shrinking during polymerization.



3-Factors related to the composite formulation:

Nonbonded microfiller particles have been found to produce significant decreases in polymerization stress by acting as stress-relieving sites within the composite.

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Acid Etching:

- A physical process that creates a microscopically rough enamel surface (enamel tags)
- first successful technique developed to bond dental materials to tooth structure
- Acid used is 37% ortho-phosphoric acid
- Sometimes referred to as conditioner

Smear Layer: When a rotary or handheld instrument is used on dentin it creates a special surface texture called a smear or smear layer that closes off the dentinal tubules. This layer is lightly adhered to the dentin surface and contains tooth cuttings, saliva, bacteria, and other surface debris

Enamel Etching

Enamel consists of organic and inorganic components. Application of 37% phosphoric acid removes about 10 microns of enamel to expose prisms of enamel rods and create the classic honeycomb effect. Acid also increases surface energy for better wetting of the enamel. Resins flow into micromechanical retentive areas. Resin tags fill microscopic holes to provide retention. Retention is about 30 MPa.

Acid etching is done for a minimum of 15 to 30 seconds. Thorough rinsing for 10 seconds removes acid and dissolved calcium phosphates.

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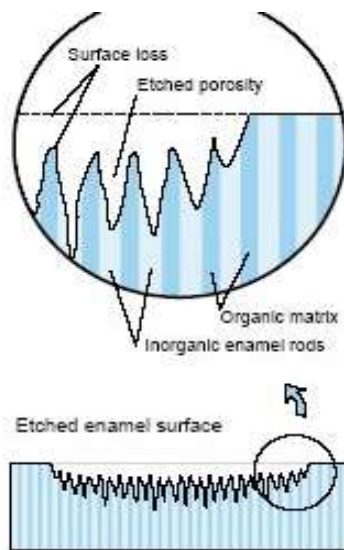


Figure 8-3. Schematic diagram depicting how acid etching produces microporosities in enamel.

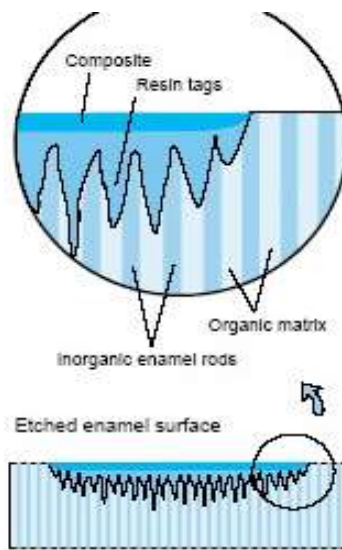


Figure 8-4. Schematic diagram depicting how resin tags penetrate the microporosities produced by acid etching of enamel.

- Over-etching results to formation of crystals (precipitates) that inhibits bonding
- Built-in quality control check – if properly etched it appears frosty or chalky white.

Dentin Etching

- 37% ortho-phosphoric acid
- Removes the smear layer from the surface of the dentin as well as the plugs of material forces into dentinal tubules during cavity preparation.
- Decalcifies a layer of dentin several microns thick. Time: 10-15 seconds.

If the etched tooth surfaces are contaminated with saliva or blood, they need to be re-etched. Such a re-etching procedure requires only 5 seconds.

Adhesion to Dentin

Conditioning or Etchant (E):- dentin etching time 15 sec only

- Removes the smear layer.
- Exposes the intertubular and peritubular collagen.
- Opens the tubules.
- Decreases the surface free energy.

Primer (P):-

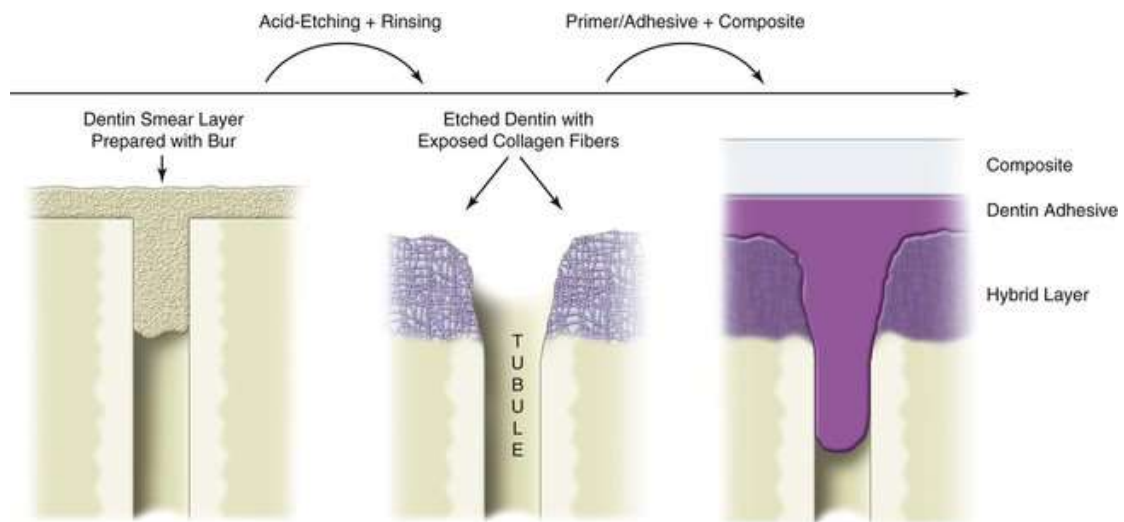
- Includes bi-functional molecules hydrophilic and hydrophobic.
- Envelops the external surface of collagen fibrils.
- Re-establishes surface free energy to levels compatible with more hydrophobic restorative materials.

Bonding Agent (B): (Adhesive resin)

- Includes monomers that are mostly hydrophobic such as Bis-GMA
- Copolymerize with the primer molecules penetrates and polymerizes into inter-fibrillar spaces to serve as a structural backbone to hybrid layer.

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Adhesion to Dentin protects the pulp, because after removal of the smear layer with Acid etch the opened dentinal tubules were sealed with a hydrophilic Primer then a bonding agent (adhesive) will be applied. During this treatment a dentin surface impregnated with resin (hybrid layer) is formed that guarantees an optimal dentin tubules closure which is acid resistant and prevents bacteria penetrating.



Current strategies for Adhesion of Resin to Dentin:

Since their introduction in 1955, dental bonding agents have evolved from **no-etch** to **total-etch** to the more recent **self-etch** systems

There are 7 generation for bonding systems the most common are:

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Total etch adhesive (etch and rinse):

In this systems Acid etch will (remove smear layer with collagen fibers will be upright and dentinal tubules opened), then primer will be applied and adhesive resin will penetrate inside opened dentinal tubules.

Total etch adhesive system Include 4th and 5th adhesive generations

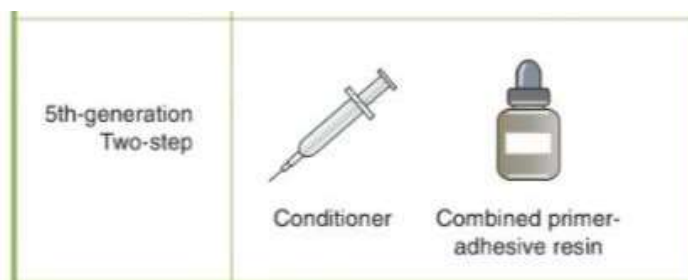
4th generation **3-steps** system:

Acid etch then Primer then Bonding



5th generation **2-step** system

Acid etch then Bonding (composed of Primer + bond resin)



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Self etch adhesive: in this system Leave smear intact, tubules plugged, partially demineralize smear layer with self- etching primer and replaced with resin filler in to tubuler dentin.

Self- etch adhesive systems include 6th and 7th generations

6th-generation 2- steps system

Self-etching primer (composed of acid etch + primer) then Bonding resin



7th generation 1- step system

Self- etching adhesive composed from (Acid etch + Primer+ Bonding resin)



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Clinical steps for acid etch and bonding for each generation

Three-Step (Fourth generation)	Two-Step (Fifth generation)	Two-Step (Sixth generation)	One-Step (Seventh generation)
1. Etch Apply for 15 seconds, rinse 15 seconds gently air dry, while keeping dentin moist.	1. Etch Apply for 15 seconds, rinse 15 seconds, gently air dry while keeping dentin moist.	1. Etch and Prime One application without rinsing, gently air-dry.	1. Etch, Prime, and Bond Apply 1 to 5 layers without rinsing, gently air-dry, light-cure.
2. Prime Apply 1 to 5 layers, gently air-dry.	2. Prime and Bond Apply 1 to 5 layers, gently air-dry, light-cure.	2. Bond Apply one layer, gently air-dry, light-cure.	
3. Bond Apply one layer, gently air-dry, light-cure.			